

Agent-Based Modeling of Decentralized Network Formation: A Distance-Based Utility Approach

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Abstract

This paper studies network formation as the outcome of decentralized strategic interactions between agents. It introduces an agent-based modeling (ABM) framework representing agents as automata with bounded information and local decision rules, allowing network structure to emerge from micro-level behavior rather than be imposed. The framework is applied to the distance-based utility model (DBUM), for which the analytical literature establishes that multiple pairwise stable networks coexist across much of the parameter space without determining which one a decentralized process reaches; we treat this as a problem of equilibrium selection. Agents optimize over a five-dimensional grid of 2,880 parameter configurations, each executed with 100 independent seeds. Resulting networks are represented in a five-dimensional feature space—density, degree centralization, clustering coefficient, largest component fraction, and global efficiency—and compared against topological benchmarks and Erdős–Rényi graphs of matched density, with pairwise stability verified directly at termination rather than inferred from convergence diagnostics.

Computational results confirm and extend analytical predictions: complete and empty networks emerge exactly where the analytical thresholds place them. In the intermediate regime, where theory leaves the outcome indeterminate, the dynamics predominantly select sparse cycle-like structures rather than the star, and lie closer to that benchmark than Erdős–Rényi graphs of equal density. Myopic radius has limited effect on the resulting topologies, indicating that strategically relevant information is concentrated in an agent’s local neighborhood. A supplementary experiment with heterogeneous link costs reverses this selection under faster decay, where cost asymmetry produces star networks through a hub-formation cascade. The fraction of runs reaching a verified pairwise stable network declines with population size, bounding the scale at which the present implementation applies.

Keywords: agent-based modeling · network formation · equilibrium selection · distance-based utility · algorithmic game theory.

1 Introduction

In the real world, the formation of networks—whether political alliances or conflict coalitions—is not something that emerges purely at random. They arise from the strategic choices of individual actors (agents) who often weigh the costs and benefits of forming or dissolving ties. There are two main aspects that define this perspective:

1. Modeling of costs and benefits
2. Prediction of outcomes based on the incentives

The canonical formalization of this perspective is the distance-based utility model (DBUM) of Jackson and Wolinsky [12], in which an agent benefits from every other agent it can reach, with benefits decaying in the length of the connecting path, and pays a fixed cost for each link it maintains. For this model the analytical literature delivers sharp characterizations: the complete network is the unique stable and efficient structure when link costs are low, the empty network prevails when costs are high, and a star encompassing all agents emerges at intermediate costs. These predictions are stated formally in Section 3 and serve as the benchmarks against which our simulation experiments are assessed.

However, the characterizations are not uniformly sharp. In the intermediate cost range, Jackson and Wolinsky establish that a star encompassing all players is pairwise stable but *not necessarily the unique* pairwise stable network. A *static equilibrium* analysis, therefore, describes the *set* of networks that could persist once reached, while remaining silent on which member of that set a decentralized process of link formation would arrive at. That question—equilibrium selection—is what motivates the present study, and it is not one that closed-form analysis of the model is equipped to answer.

A second limitation compounds the first. Richer models such as the co-author model [12] and the islands-connections model [8] generalize the DBUM by relaxing its strongest assumptions, but once heterogeneity and locality replace the uniform and global assumptions of the DBUM, closed-form solutions of stable networks become intractable. This loss of analytical tractability motivates a simulation approach in which equilibrium structures emerge endogenously from the local interactions of individual agents.

We adopt an agent-based modeling (ABM) framework for this purpose. Each agent is represented as an automaton: it holds a partial, locally-bounded view of the network, applies a fixed decision rule to that observation, and takes an action by forming or severing a link—which in turn updates the state of the network. Because no agent sees or coordinates with the full system, the resulting macro-level structure is not imposed but emerges from decentralized behavior, arising from the local decisions of each agent over time. The link-revision process we implement is that of Watts [17]: agents begin unconnected, links are severed unilaterally but formed only by mutual agreement, and agents are myopic, acting whenever a change increases their current payoff. We simulate this process across 2,880 parameter configurations spanning population size, decay, link cost, information radius, and link-proposal probability, executing each configuration with 100 independent random seeds, and we represent every resulting network in a five-dimensional feature space that permits comparison against benchmark network topologies and against Erdős–Rényi graphs of matched density.

Our contributions are the following.

1. A decentralized computational realization of the distance-based utility model in which agents evaluate link changes within a bounded information radius, corresponding to the truncated connections model of Jackson and Wolinsky [12] under the myopic link-revision process of Watts [17].
2. A systematic mapping of which structures these dynamics select across the parameter space. Watts [17] establishes that the process is path dependent and that the probability of reaching a star vanishes in the number of agents; what is reached instead has not been determined. We answer this computationally across the full grid, and find sparse, cycle-like structures throughout the intermediate cost regime.
3. A representation of simulation outcomes in a five-dimensional feature space, which allows resulting networks to be positioned relative to benchmark network topologies and to random graphs of equal density without classifying them into discrete topological types.
4. A supplementary trial in which agents are assigned heterogeneous link formation costs, showing that cost asymmetry alters which structure the same dynamics select.

We emphasize that the underlying utility function, the pairwise stability concept, and the analytical predictions against which we test are those of Jackson and Wolinsky [12], and that the link-revision dynamics are those of Watts [17]; the contribution of this work is computational rather than theoretical. For the same reason, the main experiment deliberately holds agents homogeneous: this is the setting in which analytical predictions exist, and therefore the setting in which an agent-based implementation can be validated against known results before heterogeneity is introduced. The heterogeneity trial reported in Section 5 is a first step beyond that baseline rather than a systematic treatment.

The broader aim of this line of work is to study game-theoretic network formation models and apply them to empirical studies of conflict networks, in particular to complement ongoing analytical work on the conflict network of the war in Sudan [15]. Conflict networks are a natural target for this approach: they form through decentralized, heterogeneous, and information-limited actors whose local interactions generate emergent coalition and link patterns that are hard to derive in closed form [9, 13].

The remainder of the paper is organized as follows. Section 2 situates the work within the literature on strategic network formation and agent-based modeling. Section 3 presents the formal framework and states the analytical results used as benchmarks. Section 4 describes the simulation design, the dynamics, and the convergence criterion. Section 5 reports the computational results, and Section 6 concludes.

2 Related Work

Strategic network formation. The study of networks as the outcome of deliberate individual choice builds on Myerson [16], who used graph-theoretic ideas to analyze cooperation structures in games, representing bilateral agreements between players as links and studying allocation rules that select a payoff for every such structure. Aumann and Myerson [1] treat the formation of links between players and of coalitions as endogenous. Jackson and Wolinsky [12] established the framework we adopt here: they define pairwise stability, characterize the stable and the strongly efficient networks of the connections and co-author models, and show that the two criteria need not coincide—for $N \geq 3$ there is no allocation rule that is anonymous and component balanced and that guarantees, for every value function, that at least one strongly efficient network is pairwise stable. Bala and Goyal [3] develop the principal alternative, in which the cost of a link is borne only by the agent who initiates it, so that formation can be posed as a noncooperative game; their limiting networks have simple architectures such as the wheel and the star. Jackson and van den Nouweland [10] strengthen stability to networks that are stable against changes in links by any coalition of individuals. Jackson [8] surveys this literature. What these contributions share is that they characterize the *set* of networks that can persist; they are static results about equilibria, not about the process that reaches them.

Dynamic and myopic formation. A smaller literature studies that process directly, and it is the immediate antecedent of our design. Watts [17] analyzes a dynamic model in which agents are initially unconnected and, over time, pairs of agents meet and decide whether to form or sever links: a link may be severed unilaterally, while agreement by both agents is required to form one, and agents are myopic, forming or severing a link whenever doing so increases their current payoff, with payoffs given by the connections model of Jackson and Wolinsky [12]. In each period a link is identified for updating uniformly at random. Watts shows that this process is *path dependent*, and consequently often converges to an inefficient network structure. The dynamics we simulate follow this specification, and we inherit the path dependence she establishes—a point we return to when discussing the role of the initial network. Jackson and Watts [11] develop the general theory: sequences of networks connected by myopic improvements form *improving paths*, and their Lemma 1 establishes that for any value function and allocation rule there exists at least

one pairwise stable network or closed cycle of networks. They then embed this in a stochastic process in which intended changes are accompanied by a small probability of unintended changes or errors, which selects from among the statically stable networks and cycles; networks in the support of the limiting stationary distribution of that process are termed *stochastically stable*. Their framework is the natural formal setting for the process we implement. We do not pursue that analysis here: the stochastic process induced by our agent-based implementation is treated as a computational device for reaching equilibria rather than as an object of study in its own right, and its characterization—including whether the structures we observe are stochastically stable in the sense of [11]—is left to future work.

Agent-based modeling. In agent-based modeling a system is represented as a collection of autonomous decision-making agents, each of which assesses its situation and acts on a set of rules, and which relies, as Bonabeau [4] puts it, “on the power of computers to explore dynamics out of the reach of pure mathematical methods.” Epstein and Axtell [5] established the bottom-up approach in the social sciences, and Axtell and Farmer [2] survey its use in economics and finance. Its value for the present problem is that of a laboratory. Analytical models of network formation achieve tractability in one of two ways: by imposing stronger assumptions than the phenomenon warrants—homogeneity of agents, global interaction, perfect information—or by restricting attention to those specifications for which closed-form solutions exist. Simulation is subject to neither constraint, at the cost of yielding evidence rather than proof. Bonabeau makes the first point directly: an agent-based model “is increasingly necessary as the degree of inhomogeneity increases in the modeled system.” Simulation also inverts the direction of explanation, since macro patterns of network formation are typically imposed in analytical models, whereas in agent-based modeling they emerge endogenously from the local decisions of agents.

Structural measurement. We characterize simulated networks using standard structural measures rather than topological classification. Freeman [7] develops, for each of three conceptions of centrality, measures of the centrality of positions in a network together with one “reflecting the degree of centralization of the entire network”; we use his degree-based centralization. Watts and Strogatz [18] define the clustering coefficient of a vertex as the fraction of the $k_v(k_v - 1)/2$ possible edges among its k_v neighbours that are actually present, averaged over all vertices. Latora and Marchiori [14] introduce global efficiency as a measure of how efficiently a network exchanges information. These, with density and largest-component fraction, constitute our feature space. Random graphs in the sense of Erdős and Rényi [6] provide the density-matched baseline against which the structural signature of decentralized formation is assessed.

Conflict networks. The application motivating this line of work is the formation of conflict and alliance networks. König et al. [13] combine elements from network theory with the politico-economic theory of conflict and estimate the resulting model on data from the Second Congo War, and Jackson and Nei [9] study networks of military alliances, showing how international trade bears on the density and stability of those networks. Our own analytical work on the conflict network of the war in Sudan [15] motivates the present study: the actors in such settings are heterogeneous and information-limited in ways that the assumptions securing closed-form results do not accommodate.

3 Theoretical Framework

This section fixes notation, defines the equilibrium and welfare concepts used throughout, and states the analytical results against which the simulation outcomes of Section 5 are assessed. All

results are quoted from their sources and restated in the notation of this paper; no result in this section is our own.

3.1 Networks and Payoffs

We start with a set $N = \{1, 2, \dots, n\}$ of n nodes (agents). A *graph* is a tuple (N, g) consisting of the set of nodes N and a set of edges g . An edge ij represents the link connecting agents i and j , and we write $ij \in g$ to indicate that i and j are linked under g . Such a relationship is represented by an $n \times n$ *adjacency matrix* whose entry g_{ij} equals 1 if $ij \in g$ and 0 otherwise. The notation $g + ij$ denotes the network obtained by adding link ij to g , and $g - ij$ the network obtained by deleting it. Let $G(N)$ denote the set of all undirected, unweighted networks on N , let $d(i, j)$ denote the shortest-path distance between i and j in g , and let $\deg(i)$ denote the degree of agent i . We write K_n for the complete network, C_n for the cycle, and $K_{1,n-1}$ for the star, in which a single central agent is linked to every other agent and there are no other links.

The overall benefit that a player receives from a network is modeled by a *utility function* or *payoff function*. That is, the payoff to player i is represented by a function $u_i : G(N) \rightarrow \mathbb{R}$, where $u_i(g)$ represents the net benefit that i receives from the network g . We define a *payoff structure for network g* as the vector-valued function $u : G(N) \rightarrow \mathbb{R}^n$, $g \mapsto (u_i(g))_{i \in N}$.

Let $b : \{1, \dots, n-1\} \rightarrow \mathbb{R}$ denote the benefit that the player gets from (indirect) connections as a function of the distance between the players. The *distance-based utility model* (DBUM) is one where a player's utility can be written as

$$u_i(g) = \sum_{j \neq i} [b(d(i, j))] - \deg(i) c,$$

where $d(i, j)$ is the shortest path length between i and j and $\deg(i)$ is the number of links that i has formed (its degree centrality, essentially). Let $b(k) > b(k+1) > 0$ for any k and $c \geq 0$. This embodies the idea that a player sees higher benefits for having a lower distance to other players [12].

We can elaborate further and provide one of the common structures of benefit function, $b(x) = \delta^x$, that includes the decay parameter $\delta \in (0, 1)$. We thus obtain a revised form of DBUM:

$$u_i(g) = \sum_{j \neq i} \delta^{d(i, j)} - c \deg(i),$$

with $\delta^{d(i, j)} = 0$ for agents j unreachable from i . In the *truncated* version of the model, agents benefit only from connections no more distant than a bound r , so that the sum is taken over $\{j \neq i : d(i, j) \leq r\}$. This is the specification our simulations implement, and r is the information radius of Section 4.

3.2 Stability and Efficiency

Definition 1 (Pairwise stability [12]). *A network g is pairwise stable if*

- I. *for all $ij \in g$, $u_i(g) \geq u_i(g - ij)$ and $u_j(g) \geq u_j(g - ij)$; and*
- II. *for all $ij \notin g$, if $u_i(g + ij) > u_i(g)$ then $u_j(g + ij) < u_j(g)$.*

Condition I requires that neither endpoint of an existing link wishes to sever it unilaterally; condition II requires that no absent link is one that both agents would wish to form. Together they say that, given g , no agent or pair of agents can improve by a single change to the link structure.

Definition 2 (Strong efficiency [12]). *A network g is strongly efficient relative to a profile of utility functions $(u_1, \dots, u_n)$ if $\sum_i u_i(g) \geq \sum_i u_i(g')$ for all $g' \in G(N)$.*

Definition 3 (Pareto efficiency [12]). *A network g Pareto dominates g' if $u_i(g) \geq u_i(g')$ for all i , with strict inequality for at least one i . A network g is Pareto efficient if there is no $g' \in G(N)$ that Pareto dominates it.*

The two welfare criteria are not equivalent. Strong efficiency aggregates payoffs across agents and therefore permits one agent's loss to be offset by another's gain; Pareto efficiency requires unanimity, so a single agent made worse off blocks the comparison. Consequently every strongly efficient network is Pareto efficient, while the converse fails: a network from which no reallocation of links can improve one agent without harming another may still fall short of maximizing the total. Figure 1 illustrates networks satisfying these criteria in different combinations under a single profile of utility functions, and exhibits the conflict between stability and efficiency directly: the unique pairwise stable network among those pictured is Pareto dominated by the strongly efficient one.

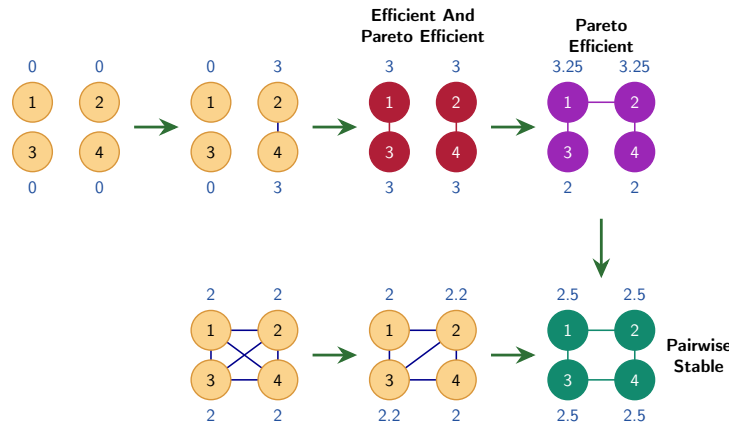


Figure 1: Networks on four agents under a single profile of utility functions, with payoffs shown beside each node. Arrows point away from networks in which some agent would benefit by deleting a link, or two agents would each benefit by adding one, and towards the network that results if those agents act. Both paths terminate at the cycle, the only pairwise stable network among those pictured, which is Pareto dominated by the strongly efficient network. Adapted from [8], Figure 6.1.

It is important to be clear about what the divergence between these criteria does and does not imply. Strong efficiency is a criterion applied by an outside observer, the researcher, to the profile of payoffs as a normative welfare criterion; no agent in the model has $\sum_i u_i$ as an objective. A pairwise stable network that is not strongly efficient is therefore not one that agents have any incentive to leave: by definition, no agent or pair can improve upon it by a single change. The divergence is a result rather than a prediction of instability, and Jackson and Wolinsky [12] show it cannot be dispensed with—for $n \geq 3$ there is no allocation rule that is anonymous and component balanced and that guarantees, for every value function, that at least one strongly efficient network is pairwise stable. Departures from a pairwise stable network require an ingredient the model does not contain, such as transfers between agents or deviations by coalitions of three or more, the latter studied by Jackson and van den Nouweland [10].

3.3 Static Benchmarks

In Jackson and Wolinsky [12], it is then proven that for some parameters of the distance-based utility model, the complete network is efficient, but not stable. For others, which have higher costs, the star network might be stable but not efficient, and sometimes, no efficient network is stable. So, we come to the conclusion that even when agents behave rationally, the equilibrium structure of the network can be inefficient. The following two propositions state these

characterizations precisely, as functions of the cost and decay parameters.

Proposition 4 (Jackson and Wolinsky [12], Prop. 1). *The unique strongly efficient network in the distance-based utility model is*

- (i) *the complete network K_n if $c < \delta - \delta^2$;*
- (ii) *a star encompassing everyone if $\delta - \delta^2 < c < \delta + \frac{n-2}{2}\delta^2$; and*
- (iii) *the empty network if $\delta + \frac{n-2}{2}\delta^2 < c$.*

Proposition 5 (Jackson and Wolinsky [12], Prop. 2). *In the distance-based utility model:*

- (i) *a pairwise stable network has at most one non-empty component;*
- (ii) *for $c < \delta - \delta^2$, the unique pairwise stable network is the complete network K_n ;*
- (iii) *for $\delta - \delta^2 < c < \delta$, a star encompassing all players is pairwise stable, but not necessarily the unique pairwise stable network; and*
- (iv) *for $\delta < c$, any non-empty pairwise stable network is one in which each agent has at least two links, and is therefore inefficient.*

Two features of Proposition 5 bear directly on the experiments reported below. Part (iii) leaves the intermediate cost regime underdetermined: the star is stable there, but so may be other networks, and the characterization is silent on which of them a formation process reaches. Part (iv) implies that in the high-cost regime the star is *not* stable, since its centre would not maintain links to agents that bring no indirect benefit, while non-empty stable networks must be at least 2-regular in the sense that every agent holds two or more links.

The truncated model is likewise covered. Jackson and Wolinsky note that Propositions 4 and 5 continue to hold when benefits are truncated at a bound r , and add the following.

Proposition 6 (Jackson and Wolinsky [12], Prop. 3). *In the truncated connections model with bound r :*

- (i) *$d(i, j) \leq 2r - 1$ for all i and j belonging to a pairwise stable component; and*
- (ii) *for $r = 2$ and $\delta < c$, no member of a pairwise stable component is in a position to disconnect all the paths connecting any two other players by unilaterally severing links.*

Part (i) bounds the diameter of a pairwise stable component as a function of the information radius, and is directly testable against simulation output.

3.4 Dynamic Benchmarks

The results above characterize which networks can persist. They do not say which of them would a decentralized formation process reach. Watts [17] addresses this for the process our simulations implement, in which agents begin unconnected, a link is selected uniformly at random in each period, links are severed unilaterally but formed only by mutual consent, and agents act myopically whenever a change increases their current payoff.

Proposition 7 (Watts [17], Prop. 3). *If $(\delta - c) > \delta^2 > 0$, then every link forms as soon as possible and remains, no link ever being broken. If $(\delta - c) < 0$, then no link ever forms.*

Proposition 8 (Watts [17], Prop. 4). *Assume $0 < (\delta - c) < \delta^2$. For $3 < n < \infty$ there is a positive probability $0 < p(\text{star}) < 1$ that the formation process converges to a star. However, as n increases $p(\text{star})$ decreases, and as n goes to infinity $p(\text{star})$ goes to 0.*

The conditions of Propositions 7 and 8 partition the parameter space in exactly the way Proposition 5 does: $(\delta - c) > \delta^2$ is $c < \delta(1 - \delta)$, the regime in which the complete network is uniquely stable; $0 < (\delta - c) < \delta^2$ is $\delta(1 - \delta) < c < \delta$, the intermediate regime; and $(\delta - c) < 0$ is $c > \delta$. Watts therefore supplies the dynamic counterpart to the static characterization in the two extreme regimes, and in the intermediate regime she shows that the star, though stable, is reached with probability tending to zero in the number of agents. The mechanism is her Lemma 1: if links form between two disjoint pairs of agents, the star can never subsequently form, and the probability that the next pair drawn is disjoint from a given pair is $(n - 2)(n - 3)/n(n - 1)$, which approaches 1 as n grows.

Remark 9. Watts’s notion of stability extends pairwise stability by allowing an agent to form a link and simultaneously sever others; she notes that Jackson and Wolinsky’s proofs adapt to this setting. Our implementation admits one action per step and therefore corresponds to pairwise stability as defined above.

4 Methodology

This section presents the computational experiment designed to validate and extend the analytical predictions of the distance-based utility model through agent-based simulation. The goal of the experiment is to examine how network structures emerge endogenously as agents locally optimize their utilities under varying decay, cost parameters, myopic radii, and population sizes.¹

4.1 Model Setup

We consider populations of $n \in \{10, 20, \dots, 80\}$ agents represented as nodes in the network g , initially empty. For each population size, agents form or sever links over time according to a myopic, utility-increasing rule. Time is discrete, and at each step a single agent is selected uniformly at random to evaluate a potential local network modification. Agent preferences are represented by a distance-based utility function, where benefits decay exponentially with graph distance and each direct link has a fixed cost. Agents are assumed to only evaluate marginal changes to their current network position.

4.2 Distance-Based Utility Function

For agent i , utility is defined as

$$u_i(g) = \sum_{j \neq i, d(i,j) \leq r} \delta^{d(i,j)} - c \cdot \deg(i),$$

where $d(i, j)$ is the shortest-path distance between agents i and j , $\delta \in (0, 1)$ is the decay parameter, $c > 0$ is the per-link maintenance cost, and $\deg(i)$ is the degree centrality of agent i . Agent utility is evaluated within a fixed radius r , ensuring myopic behavior. Only nodes with $d(i, j) \leq r$ contribute to the benefit term. This is the truncated connections model of Jackson and Wolinsky [12], with r playing the role of their truncation bound; Propositions 4 and 5 continue to hold under this specification, and Proposition 6 additionally bounds the diameter of a pairwise stable component by $2r - 1$.

4.3 Agent-Based Dynamics

At each iteration, the algorithm performs the following steps:

1. Select an agent i uniformly at random.

¹Simulation code available at <https://github.com/DivergenceTheorem/abm-network-dbum>.

2. With probability $p_{\text{add}} \in \{0.25, 0.50, 0.75\}$, attempt to add a link; otherwise, attempt to remove an existing link.
3. If adding a link, select a non-neighbor j uniformly at random.
4. If removing a link, select an existing neighbor j uniformly at random.
5. Compute the marginal utility change Δu_i induced by the proposed action.
6. Implement the action if $\Delta u_i > 0$. In the case of link addition, the link is formed only if both $\Delta u_i > 0$ and $\Delta u_j > 0$, reflecting the bilateral nature of link formation under pairwise stability.

This is the link-revision process studied by Watts [17]: agents begin unconnected, a link may be severed unilaterally but is formed only by mutual consent, and agents act myopically on the current payoff. Two details differ. Watts selects a link uniformly at random in each period, whereas we select an agent uniformly and then a candidate partner; and her formulation allows an agent to form a link while simultaneously severing others, whereas we admit one action per step, so that our resting points are pairwise stable in the sense of Definition 1.

Because a move is accepted only when it strictly increases the utility of every agent whose consent it requires, the process is a better-response rather than a best-response dynamic, in the sense distinguished by Jackson and Watts [11]: an agent evaluates one sampled alteration, not the full set of available ones. It follows that any network at which the process comes to rest is pairwise stable, since no admissible change would be accepted there. It does not follow that every run ends at such a network: a run halted by the stopping rule of Subsection 4.5 or by the step cap may not have exhausted the available improvements. We therefore verify pairwise stability directly at termination and report the fraction of runs that satisfy it.

4.4 Network Formation Algorithm

Algorithm 1 Distance-Based Utility Network Formation

```

1: Initialize graph  $(N, g)$  with  $n$  isolated nodes;  $t \leftarrow 0$ 
2: while not converged and  $t < t_{\text{max}}$  do
3:    $t \leftarrow t + 1$ 
4:   Select agent  $i \sim \text{Uniform}(\{1, \dots, n\})$ 
5:   if  $\text{Uniform}(0, 1) < p_{\text{add}}$  then ▷ Attempt link addition
6:     Select  $j$  uniformly from non-neighbors of  $i$ 
7:     if  $\Delta u_i^{\text{add}}(i, j) > 0$  and  $\Delta u_j^{\text{add}}(i, j) > 0$  then
8:       Add edge  $(i, j)$  to  $g$ 
9:     end if
10:  else ▷ Attempt link removal
11:    Select  $j$  uniformly from neighbors of  $i$ 
12:    if  $\Delta u_i^{\text{remove}}(i, j) > 0$  then
13:      Remove edge  $(i, j)$  from  $g$ 
14:    end if
15:  end if
16:  if  $t \bmod \tau = 0$  then ▷ Check convergence every  $\tau$  steps
17:    Update convergence state (see Algorithm 2)
18:  end if
19: end while

```

4.5 Convergence Criterion and Stopping Rule

Rather than running simulations for a fixed number of steps, convergence is detected using a rolling window criterion over the five-dimensional feature vector of the network (defined in Subsection 4.7).

Every $\tau = 10$ steps a convergence check is performed, and let $s = 1, 2, 3, \dots$ be the indices of these checks. At check s , the current feature vector $\mathbf{m}(G_t)$ is appended to a fixed-size deque $\mathcal{H}$ of length $W = 10$, and the window mean is defined as

$$\mu_s = \frac{1}{W} \sum_{i=0}^{W-1} \mathbf{m}(G_{t-i\tau}) \in \mathbb{R}^5.$$

The vector μ_s has five coordinates

$$\mu_s = (\mu_{s,1}, \mu_{s,2}, \mu_{s,3}, \mu_{s,4}, \mu_{s,5}),$$

corresponding respectively to the window-averaged density, degree centralization, average clustering coefficient, largest component fraction, and global efficiency.

Consecutive window means are compared using the ℓ^∞ metric:

$$\|\mu_s - \mu_{s-1}\|_\infty = \max_{k \in \{1, \dots, 5\}} |\mu_{s,k} - \mu_{s-1,k}|.$$

A window comparison is declared stable if $\|\mu_s - \mu_{s-1}\|_\infty < \varepsilon$, with $\varepsilon = 0.0005$. The simulation terminates once $K = 5$ consecutive stable comparisons are recorded. To avoid false early termination, the first $B = 10$ checks are skipped as a mandatory burn-in period. A hard cap of $t_{\max} = 20 \cdot \frac{n(n-1)}{2}$ steps prevents any single run from exceeding a quadratic multiple of the number of possible edges. The window size W and burn-in count B are held constant across all population sizes.

Two properties of this rule should be made explicit. First, $\mathbf{m}(\cdot)$ is a summary statistic rather than a graph invariant: distinct networks can share the same five coordinates, so stabilization of μ_s indicates that the aggregate structure has settled, not that the graph itself has stopped changing. The criterion is therefore a stopping rule, and what certifies stability is the direct verification described below. Second, ε is an absolute threshold applied to quantities whose granularity depends on n . A single edge alters density by $2/n(n-1)$ and, because consecutive windows overlap in $W-1$ of their entries, shifts the window mean by a further factor of W . At $n = 10$ one edge change is enough to exceed ε ; at $n = 80$ roughly sixteen edges must change within a window before the criterion registers movement. The rule is thus sharper for small populations than for large ones, which bears directly on the convergence rates reported in Section 5.

Algorithm 2 Rolling Window Convergence Check

Require: Deque $\mathcal{H}$ of capacity W ; previous mean μ_{s-1} ; stable count q ; check count b ; parameters ε, K, B, W

```

1:  $b \leftarrow b + 1$ ; append  $\mathbf{m}(G_t)$  to  $\mathcal{H}$ 
2: if  $b > B$  and  $|\mathcal{H}| = W$  then
3:    $\mu_s \leftarrow \frac{1}{W} \sum_{\mathbf{v} \in \mathcal{H}} \mathbf{v}$ 
4:   if  $\mu_{s-1}$  is defined then
5:      $\Delta \leftarrow \|\mu_s - \mu_{s-1}\|_\infty = \max_k |\mu_{s,k} - \mu_{s-1,k}|$ 
6:      $q \leftarrow q + 1$  if  $\Delta < \varepsilon$ , else  $q \leftarrow 0$ 
7:     if  $q \geq K$  then return CONVERGED
8:   end if
9: end if
10:  $\mu_{s-1} \leftarrow \mu_s$ 
11: end if
12: return NOT CONVERGED

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4.6 Parameter Grid and Experimental Design

Simulations are run over a five-dimensional grid of $(n, \delta, c, r, p_{\text{add}})$ parameter tuples. For each parameter setting:

- The network is reset to an empty graph with n isolated nodes
- The simulation runs until convergence or until the hard cap is reached
- The resulting network structure is recorded

Each configuration is executed with $S = 100$ independent random seeds. Each seed initializes an empty graph and runs the dynamics independently. The same S seeds are used at every parameter configuration, so that comparisons across δ , c , r and p_{add} are paired rather than independent. Across the 100 runs, we compute the mean, standard deviation, and 95% confidence interval $(1.96 \cdot \sigma / \sqrt{S})$ for all structural metrics and benchmark distances. In addition, pairwise stability is verified directly at termination of each run, and we report the fraction of seeds that converged to a pairwise stable network (`PS_frac`). This fraction is computed over simulated runs only; configurations resolved analytically, as described in the remark below, are excluded from it.

The population size takes values $n \in \{10, 20, \dots, 80\}$ (eight values), the decay parameter $\delta \in \{0.35, 0.65, 0.95\}$, the cost parameter $c \in \{0.1, 0.2, \dots, 1.0\}$ (ten values), the myopic radius $r \in \{3, 5, 8, 10\}$, and the link addition probability $p_{\text{add}} \in \{0.25, 0.5, 0.75\}$. This yields $8 \times 3 \times 10 \times 4 \times 3 = 2,880$ parameter configurations in total. Population sizes beyond $n = 80$ were not simulated, as the cost of running 100 seeds at every configuration becomes prohibitive; the consequences of this bound are discussed in Section 6.

Remark 10. For any setting where $\delta \leq c$, the marginal utility of forming a direct link is non-positive regardless of the network state. In such cases the simulation is bypassed and the empty graph is recorded directly, without running any dynamics. For $\delta < c$ this is exactly the second clause of Proposition 7, which states that no link ever forms; the boundary case $\delta = c$ follows from the requirement that an accepted move strictly increase utility.

4.7 Network Representation and Benchmarking

Rather than classifying networks into discrete topological types, we represent each resulting network in a five-dimensional feature space. For a graph G on n nodes, the feature vector is defined as

$$\mathbf{m}(G) = (\rho(G), C(G), \bar{\kappa}(G), \lambda(G), E(G)) \in [0, 1]^5,$$

where

- $\rho(G) = \frac{2|E|}{n(n-1)}$ network density,
- $C(G) = \frac{\sum_i (d_{\text{max}} - d_i)}{(n-1)(n-2)}$ degree centralization in the sense of Freeman [7],
- $\bar{\kappa}(G) = \frac{1}{n} \sum_{i=1}^n \kappa_i$ average local clustering coefficient [18], where the local clustering coefficient of node i is

$$\kappa_i = \begin{cases} \frac{|\{e_{jk} : j, k \in \mathcal{N}_i, e_{jk} \in E\}|}{d_i(d_i - 1)/2} & \text{if } d_i \geq 2, \\ 0 & \text{otherwise,} \end{cases}$$

where $\mathcal{N}_i$ denotes the neighborhood of i and $d_i = |\mathcal{N}_i|$ its degree,

- $\lambda(G) = \frac{|\mathcal{C}_1|}{n}$ fraction of nodes in the largest connected component $\mathcal{C}_1$, with isolated nodes counted as components of size one,
- $E(G) = \frac{1}{n(n-1)} \sum_{i \neq j} \frac{1}{d(i,j)}$ global efficiency in the sense of Latora and Marchiori [14], with $1/\infty = 0$ for disconnected pairs.

To quantify the structural similarity between any two networks, we use a normalized Euclidean distance in the feature space:

$$d(\mathbf{m}_A, \mathbf{m}_B) = \sqrt{\frac{1}{5} \sum_{k=1}^5 (m_{A,k} - m_{B,k})^2}.$$

Four benchmark configurations are defined for each population size n :

- $G^\emptyset$ —the empty graph (no edges),
- G° —the cycle graph C_n (2-regular),
- $G^\star$ —the star graph $K_{1,n-1}$,
- G^K —the complete graph K_n .

These benchmarks correspond to the resulting topologies predicted by the analytical literature [12]: the empty network emerges at high cost, the complete network at low cost and slow decay, and the star at moderate cost with slow decay.

Because $\mathbf{m}(\cdot)$ summarizes a network in five numbers, networks that differ structurally may lie close together in this space, and the coordinates are not equally informative for every comparison. The cycle and the star are a case in point: both are sparse, connected and free of triangles, so they differ appreciably only in degree centralization and global efficiency, and the first of these accounts for the greater part of the separation between them. Distances reported below should be read accordingly.

For each ABM simulation outcome G , the distance to each benchmark is computed as

$$d_b(G) = d(\mathbf{m}(G), \mathbf{m}(G^b)), \quad b \in \{\emptyset, \circ, \star, K\}.$$

To assess whether the resulting structure differs from a random network at the same density, we construct an Erdős–Rényi (ER) baseline [6]. For each ABM outcome with density $\rho(G)$, we generate $M = 100$ independent Erdős–Rényi graphs $G_1^{\text{ER}}, \dots, G_M^{\text{ER}} \sim \mathcal{G}(n, \rho(G))$ with fixed seeds and compute their average feature vector:

$$\mu^{\text{ER}}(\rho) = \frac{1}{M} \sum_{m=1}^M \mathbf{m}(G_m^{\text{ER}}).$$

The distance from ER graphs to benchmarks is computed in the same manner. Comparing distances from ER graphs and ABM networks to benchmarks determines whether ABM-generated networks are structurally closer to or farther from a given network archetype relative to a random graph of the same density.

Remark 11. The ER comparison is in expectation: the generated graphs have edge probability $\rho(G)$ rather than exactly the density of G , and their metrics are averaged over the M replicates.

4.8 Heterogeneity Trial

As a proof-of-concept, we conduct an experiment in which agents are equipped with heterogeneous link formation costs.

For a fixed population of $n = 10$ agents, the ten cost values $\{0.1, 0.2, \dots, 1.0\}$ are assigned bijectively to the ten agents, so that each agent i carries a distinct cost c_i and every value is used exactly once. The assignment is drawn afresh as a uniform random permutation for each seed, so that results are averaged over which agent holds which cost rather than conditioned on a single labelling.

The parameter grid reduces to $\delta \in \{0.35, 0.65, 0.95\}$ and $r \in \{3, 5, 7, 10\}$, giving 12 configurations, with $p_{\text{add}} = 0.5$ fixed. Apart from the radius values, which differ from the main experiment in their third entry, and the substitution of agent-specific costs for a common one, the design is unchanged: the same dynamics, convergence criterion, seed count and benchmark comparisons apply. Note that no configuration can be resolved analytically here, since there is no single cost to compare against δ ; every run is simulated. This trial is intended to test whether the findings of the experiment with homogeneous agents persist when agents differ in how expensive it is for them to form links.

5 Results and Discussion

We present simulation outcomes and discuss how variation in the model’s parameters affects the dynamics of network formation and the resulting topologies.

5.1 Confirmation of Analytically Predicted Structures

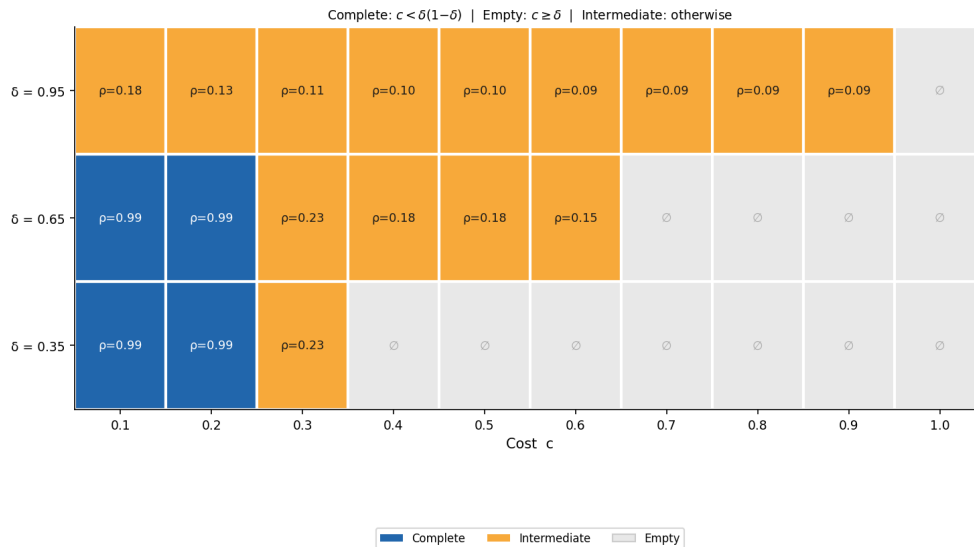


Figure 2: Resulting network densities averaged over myopic radius, agent populations, and link addition probability

From experimental data, the only configurations under which complete networks emerged were $\delta \in \{0.35, 0.65\}$ at cost values $c \in \{0.1, 0.2\}$ —it confirms the analytical prediction of [12] that the complete network K_n is the unique pairwise stable network when $c < \delta(1 - \delta)$ (Proposition 5(ii)). It is worth noting that $c < \delta(1 - \delta)$ is symmetric around $\delta = 0.5$, so both $\delta = 0.35$ and $\delta = 0.65$ yield the same threshold of $0.35 \times 0.65 = 0.2275$. The parameter grid satisfies $c < 0.2275$ precisely at $c \in \{0.1, 0.2\}$, and complete networks are observed exactly at those configurations. For $\delta = 0.95$ the threshold falls to 0.0475 which is below the minimum tested cost

of $c = 0.1$, so no complete networks were expected or observed (Figure 2). The corresponding dynamic prediction is Proposition 7: in this cost range every link forms and none is ever broken.

When $c \geq \delta$, the marginal utility of any direct link $\delta - c \leq 0$, so no agent has an incentive to form a link regardless of network state, and by Proposition 7 no link ever forms. The empty network is therefore the outcome of the dynamics from an empty initial network. It is not the only pairwise stable network in this region: Proposition 5(iv) admits non-empty pairwise stable networks in which every agent holds at least two links, but such networks are not reachable from an empty start. This condition is met in 1,152 of the 2,880 configurations, which are recorded analytically without running the dynamics.

According to Proposition 5(iii), for $\delta(1-\delta) < c < \delta$ a star encompassing all players is pairwise stable, though for some n and parameter values in this range it is not the unique pairwise stable network. In our experiment, none of the parameter combinations produced a star network. This is what Proposition 8 predicts for these dynamics: the probability that the formation process converges to a star decreases in the number of agents and tends to zero. The mechanism is that once links have formed between two disjoint pairs of agents the star can no longer assemble, and the probability that the next pair drawn is disjoint from a given pair is $(n-2)(n-3)/n(n-1)$, already 0.62 at our smallest population of $n = 10$. Above $c = \delta$ the star is in any case not stable, since by Proposition 5(iv) every agent in a non-empty pairwise stable network must hold at least two links, which the centre of a star does not.

Pairwise stability was verified directly at the termination of every run, by checking each existing link against condition I and each absent link against condition II of Definition 1. Table 1 reports the fraction of runs satisfying both, computed over the 216 simulated configurations at each population size; the 144 configurations per population size that are resolved analytically are excluded, since no dynamics are run for them.

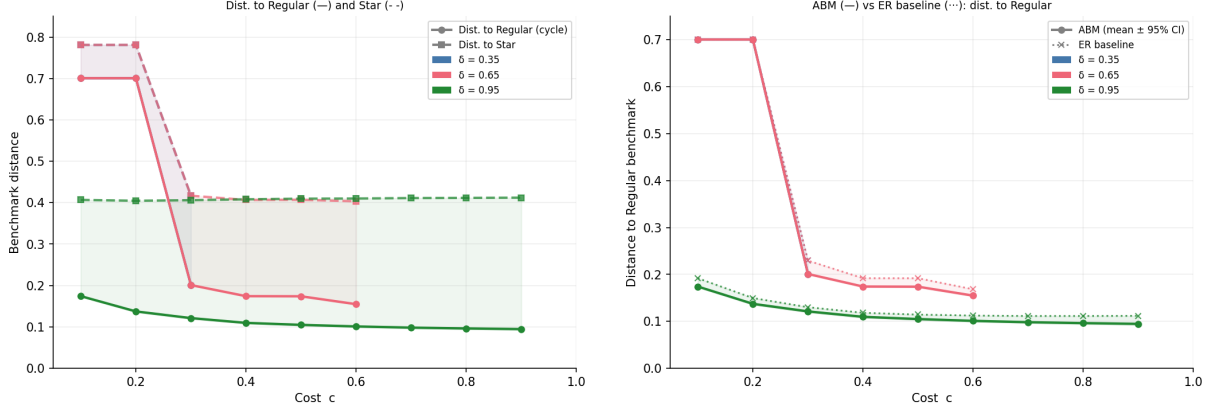
N	Configurations	Pairwise stable fraction
10	216	0.911
20	216	0.375
30	216	0.207
40	216	0.161
50	216	0.107
60	216	0.057
70	216	0.027
80	216	0.006

Table 1: Fraction of runs terminating at a verified pairwise stable network, by population size, averaged over the simulated configurations at each N .

The fraction declines sharply with population size. Two explanations are consistent with this pattern, and the present experiment does not distinguish between them. The process may be in a closed cycle of networks, which Lemma 1 of [11] admits as the alternative to a pairwise stable network at the end of an improving path; or the run may have terminated, either through the stopping rule or the step cap, before the available improvements were exhausted. Distinguishing the two requires tracking the trajectory rather than its endpoint, and is left to future work. The structural findings reported below concern the terminal networks themselves and do not depend on which explanation holds.

5.2 Shift towards 2-Regular Networks

As the δ parameter increases, resulting ABM networks shift towards more cycle-like topologies. The clearest signal is the distance to the 2-regular (cycle) benchmark: it falls from 0.701 in the complete network configuration to within 0.121–0.174 at $\delta = 0.95$ in the lower cost range



(a) Distance to 2-regular and star benchmarks (non-empty runs, averaged over myopic radius and agent populations)

(b) Comparison of distance to 2-regular benchmark between ABM and ER networks (non-empty runs, averaged over myopic radius and agent populations)

Figure 3: Benchmark distances across the parameter grid.

(roughly a fourth of its original value). For intermediate cost regimes, the decrease in distance is moderate: for $\delta \in \{0.35, 0.65\}$ the distance to the 2-regular benchmark is within 0.101–0.174 range; for $\delta = 0.95$, the distance range is 0.101–0.110. The distance to the star benchmark also decreases with δ , but far less dramatically, from 0.781 to 0.406 at lower costs, staying within the 0.403–0.410 range for moderate costs, essentially flattening out.

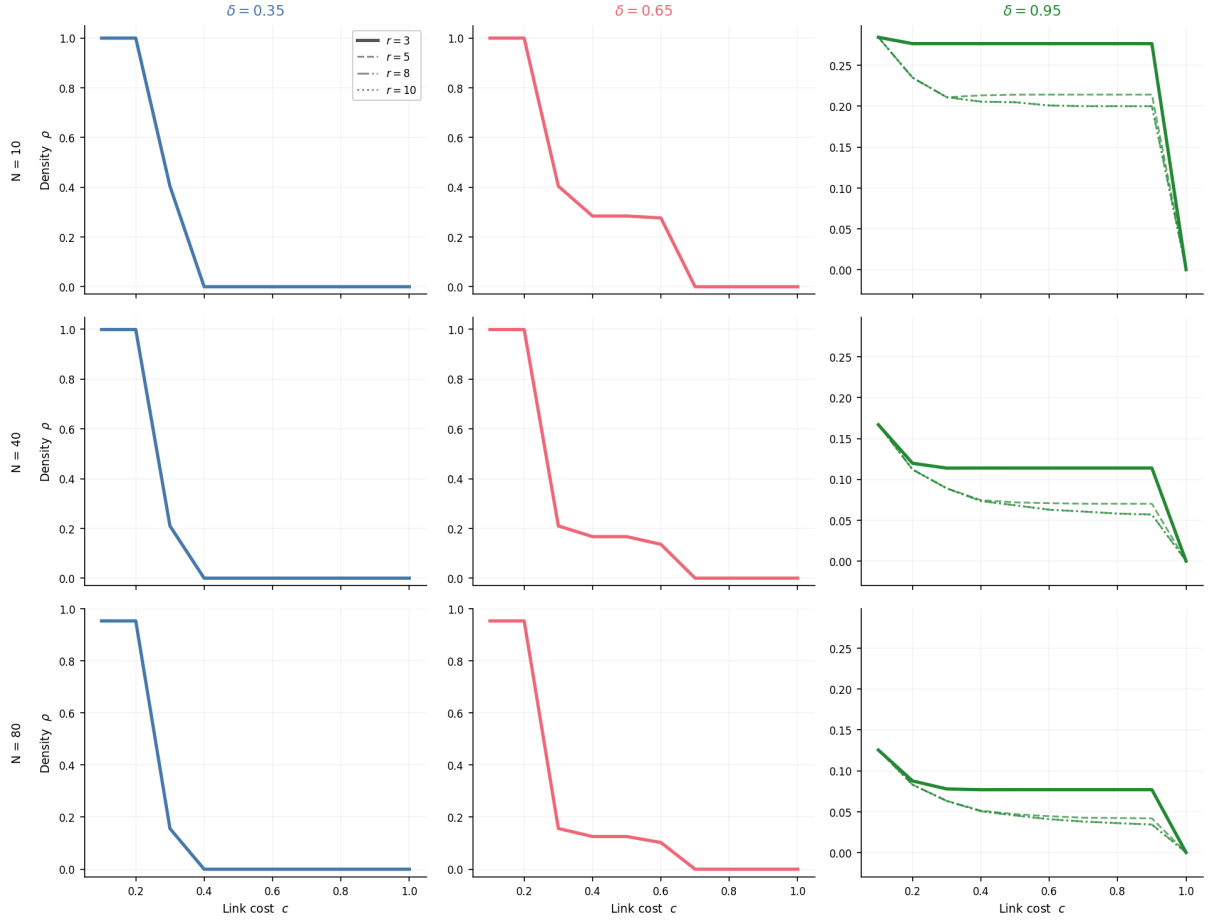
Across all non-empty parameter configurations, the 2-regular benchmark was the closest of the four in every one of the 864 configurations at $\delta = 0.95$, in 66.6% of the 576 at $\delta = 0.65$, and in 33.3% of the 288 at $\delta = 0.35$ —the last figure reflecting that two-thirds of the configurations at $\delta = 0.35$ are complete networks, as expected analytically. The 2-regular benchmark, rather than the star, is therefore the structure most frequently selected by these dynamics (Figure 3a).

For parameters $\delta \in \{0.35, 0.65\}$, once the cost threshold $c = \delta(1 - \delta) = 0.2275$ is passed and complete network is no longer a unique pairwise stable configuration, the distance to 2-regular and star network decreases rapidly. For $\delta = 0.95$, whose threshold lies at $c = 0.0475$ below the minimum tested cost, networks are sparse and cycle-like across the entire parameter grid.

Relative to the ER random graph of matched density, ABM networks are consistently closer to the 2-regular benchmark at every parameter configuration that does not result in the empty network. The only exception is the complete network regime ($c \in \{0.1, 0.2\}$ for $\delta \in \{0.35, 0.65\}$), where both ABM and ER yield the same density $\rho \approx 1$ and are therefore at equal distance from all benchmarks (Figure 3b). It is worth noting that the distance gap is more explicit for lesser agent populations: when we average for $N \in \{10, 20\}$, we obtain a difference range of 0.040–0.081, while averaging over $N \geq 30$ yields range of 0.001–0.011, with a few instances (3 out of 30 combinations of δ and c) showing ER actually being closer to the regular network (by at most 0.0031). This is consistent with the decline in verified pairwise stability reported in Table 1: at larger populations fewer runs terminate at a network from which no improving move remains.

5.3 Radius Insensitivity

For parameters $\delta \in \{0.35, 0.65\}$, the four tested radii produce identical networks at termination: density and benchmark distances are indistinguishable across $r \in \{3, 5, 8, 10\}$ (Table 2 and Table 3) with slight increase for $\delta = 0.95$ as demonstrated in Table 3 and Figure 4.

Figure 4: Network density vs. cost over different δ , N , and r parameters

δ	Benchmark	Mean $\pm$ Std	95% CI	Worst
0.35	Empty	0.0000 $\pm$ 0.0000	[0.0000, 0.0000]	0.0000
	Complete	0.0000 $\pm$ 0.0000	[0.0000, 0.0000]	0.0000
	Star	0.0000 $\pm$ 0.0000	[0.0000, 0.0000]	0.0000
	Regular	0.0000 $\pm$ 0.0000	[0.0000, 0.0000]	0.0000
0.65	Empty	0.0001 $\pm$ 0.0002	[0.0001, 0.0001]	0.0009
	Complete	0.0004 $\pm$ 0.0006	[0.0003, 0.0005]	0.0027
	Star	0.0003 $\pm$ 0.0006	[0.0002, 0.0004]	0.0049
	Regular	0.0002 $\pm$ 0.0004	[0.0001, 0.0003]	0.0022
0.95	Empty	0.0193 $\pm$ 0.0110	[0.0178, 0.0207]	0.0359
	Complete	0.0328 $\pm$ 0.0179	[0.0304, 0.0351]	0.0568
	Star	0.0050 $\pm$ 0.0042	[0.0044, 0.0055]	0.0163
	Regular	0.0380 $\pm$ 0.0257	[0.0346, 0.0415]	0.0760

Table 2: Max-min spread of benchmark distances across $r \in \{3, 5, 8, 10\}$ for each fixed configuration $(\delta, N, c, p_{\text{add}})$. The 95% CI is on the mean spread across configurations.

δ	Mean $\pm$ Std	95% CI	Worst
0.35	0.0000 $\pm$ 0.0000	[0.0000, 0.0000]	0.0000
0.65	0.0003 $\pm$ 0.0005	[0.0002, 0.0004]	0.0024
0.95	0.0390 $\pm$ 0.0235	[0.0359, 0.0422]	0.0791

Table 3: Max-min spread of network density across $r \in \{3, 5, 8, 10\}$ for each fixed configuration $(\delta, N, c, p_{\text{add}})$. The 95% CI is on the mean spread across configurations.

The exception is observed at $r = 3$ under slow decay ($\delta = 0.95$). In this setting, paths of length 4–5 would contribute utility $\delta^4 = 0.81$ and $\delta^5 = 0.77$ —values comparable to a direct link under faster decay—but at $r = 3$ they contribute nothing, since benefits are truncated at that distance. Direct links are correspondingly more valuable relative to the longer-path alternatives that would otherwise substitute for them. This drives the formation of denser terminal networks: $r = 3$ produces density that is 0.025–0.073 higher than for $r \geq 5$ across the moderate cost range, for all tested values of N . As N grows, the density gap becomes less visible.

5.4 Heterogeneity trial

Interestingly, the star network emerged as the dominant outcome for networks at $\delta \in \{0.35, 0.65\}$, while networks at $\delta = 0.95$ remained closer to the 2-regular benchmark (Table 4). One possible explanation of convergence to the star network at $\delta \in \{0.35, 0.65\}$ is that the lower-cost agents have stronger incentives to accumulate links from the start thus creating a hub that sequentially draws in agents with higher link costs. For example, at $\delta = 0.35$, only three agents with $c \in \{0.1, 0.2, 0.3\}$ find it profitable to form links and they initially form a small hub. Once the emerging hub has k spokes (two, in this case), the benefit for any outside agent i of joining the hub rises to $\delta + k\delta^2$, because connecting to the hub now provides not only a direct link but also distance-two access to all k existing members.

The benefit of joining the hub grows with each new spoke: at $k = 2$ it reaches 0.595, drawing in agents with $c \in \{0.4, 0.5\}$; at $k = 4$ it reaches 0.840, drawing in $c \in \{0.6, 0.7, 0.8\}$; and at $k = 6$ it exceeds 1.0, making hub membership rational even for the highest-cost agent. Simultaneously, the direct links between non-hub agents become unprofitable once the hub provides an intermediary: the marginal benefit of a peer link drops from δ to $\delta - \delta^2 = \delta(1 - \delta)$, which falls below most agents’ link costs. The decentralized process therefore reinforces hub-spoke connections and prevents direct links, assembling the star network by consecutive decentralized evolution.

At $\delta = 0.95$, by contrast, $\delta^2 \approx 0.90 \approx \delta$, so distance-two access is nearly as valuable as a direct link and the structural advantage of the hub role disappears. A 2-regular topology, where every agent maintains exactly two links at equal cost, becomes the decentralized outcome instead.

The persistent convergence to the sparse network topologies explains the low variability in density that stayed within the 0.20–0.21 range for all configurations except at $r = 3$, $\delta = 0.95$, where the density reached 0.268. This is consistent with the earlier finding that a small radius combined with low decay led to denser networks in the homogeneous agent configurations. The radius insensitivity was confirmed across all parameters except for the same combination of $r = 3$, $\delta = 0.95$.

δ	r	Empty	Complete	Star	Regular
0.35	3	0.635 ± 0.046	0.718 ± 0.046	0.051 ± 0.078	0.403 ± 0.075
	5	0.635 ± 0.046	0.719 ± 0.046	0.050 ± 0.078	0.403 ± 0.075
	7	0.635 ± 0.046	0.719 ± 0.046	0.050 ± 0.078	0.403 ± 0.075
	10	0.635 ± 0.046	0.719 ± 0.046	0.050 ± 0.078	0.403 ± 0.075
0.65	3	0.646 ± 0.047	0.727 ± 0.052	0.038 ± 0.093	0.416 ± 0.088
	5	0.639 ± 0.054	0.720 ± 0.058	0.051 ± 0.104	0.403 ± 0.099
	7	0.639 ± 0.054	0.720 ± 0.058	0.051 ± 0.104	0.403 ± 0.099
	10	0.639 ± 0.054	0.720 ± 0.058	0.051 ± 0.104	0.403 ± 0.099
0.95	3	0.511 ± 0.012	0.596 ± 0.020	0.340 ± 0.055	0.125 ± 0.045
	5	0.489 ± 0.017	0.626 ± 0.015	0.330 ± 0.055	0.122 ± 0.054
	7	0.485 ± 0.019	0.632 ± 0.011	0.325 ± 0.053	0.127 ± 0.052
	10	0.485 ± 0.020	0.632 ± 0.010	0.325 ± 0.053	0.127 ± 0.052

Table 4: Mean $\pm$ standard deviation of benchmark distances for each heterogeneity trial configuration

6 Conclusion

In conclusion, this project examined the formation of networks through the lens of game-theoretic models and agent-based simulations. We began by first establishing the foundational concepts of network theory and introduced the distance-based utility model as a tractable framework for demonstrating how agents weigh the costs and benefits of both direct and indirect connections.

The computational experiment demonstrated that agent-based simulation can successfully reproduce and extend analytical predictions of the distance-based utility model.

The analytical predictions of resulting networks hold for a decentralized network formation model—the complete networks emerged exactly at $c < \delta(1 - \delta)$, empty networks at $c \geq \delta$. Within the moderate range, ABM dynamics lean towards cycle-like structures rather than the star networks. The decay parameter δ is the primary driver: as δ increases, resulting networks shift toward the 2-regular benchmark, consistently closer to the 2-regular benchmark than ER graphs of matched density. The information radius has limited effect on the network formation dynamics; the exception is $r = 3$ under slow decay ($\delta = 0.95$).

That cycle-like structures rather than stars should be selected is not itself unanticipated: Proposition 8 establishes that the probability of reaching a star vanishes as the number of agents grows. What static equilibrium analysis leaves open is which of the remaining pairwise stable networks a decentralized process arrives at, and this the simulation answers across the parameter grid—sparse, cycle-like structures throughout the intermediate cost regime, and measurably closer to that benchmark than random graphs of the same density.

A particularly interesting finding emerges from our heterogeneity experiment, in which each agent is assigned a unique link formation cost drawn from $\{0.1, 0.2, \dots, 1.0\}$. Unlike the homogeneous cost setting where cycle-like structures dominate the moderate cost range, heterogeneous costs consistently produce the star network as the dominant outcome: the lower-cost agents accumulate connections early on, sequentially drawing in higher-cost agents as the hub grows large enough to make membership rational for them. This cascade logic may shed light on our empirical work on armed conflicts: hostilities often cluster around a small set of central actors for whom the engagement cost is low relative to their size, resource base, or strategic capacity. A great power or a large military coalition, by analogy, faces a structurally lower link formation cost, and the heterogeneous-cost dynamics suggest that this asymmetry alone—without any coordination—can be sufficient to generate hub-like conflict patterns from purely decentralized incentives.

A key limitation of the current study is its computational scope. The experiment covers $N \in \{10, 20, \dots, 80\}$, and the fraction of runs terminating at a verified pairwise stable network

falls steeply with population size, from 0.911 at $N = 10$ to 0.006 at $N = 80$ (Table 1).

Larger populations require substantially more steps to explore the link space, and the terminal networks at those sizes are either resting in a closed cycle, in the sense of Lemma 1 of [11], or were reached before the available improvements had been exhausted; the present experiments do not distinguish between the two.

Whether the same structures obtain at substantially larger populations is not settled here. The mechanism suggests they should: the selection of a cycle-like structure follows from a local comparison between the value of a direct link and the indirect benefit of an alternative path, and that comparison does not obviously scale with the size of the population.

A second limitation concerns the initial network. Every run begins from the empty graph, and since the process is path dependent [17], the structures reported here are those reachable from that starting point; characterizing the outcomes reachable from other initial networks is left to future work.

The broader goal of this project is to study more advanced game theoretic network formation models and apply them to empirical studies of conflict networks to reveal insights about agents' incentives and possibly reveal conflict evolution dynamics that are not tractable by analytical methods.

As a first step, we intend to expand on the heterogeneous cost finding described above, to explore convergence criteria whose sensitivity does not degrade with population size, and to push the simulations to larger agent populations and more complex heterogeneous setups.

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